1. Create user with name Techie and provide sudo access

Command Executed:

sudo useradd khadeer

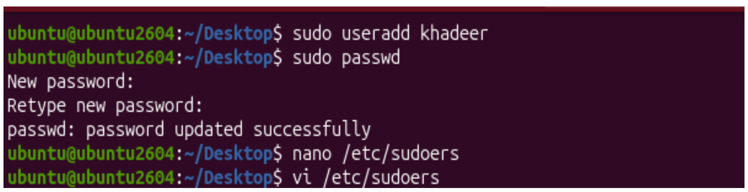
sudo passwd khadeer@

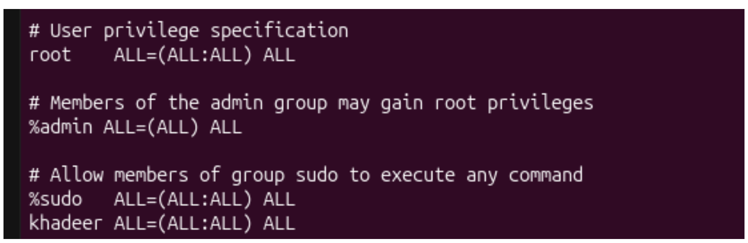
sudo su khadeer

vi /etc/sudoers

Command Output:
User created successfully, password set, and user added to sudo group.

Evidence (Screenshot):





Key Learning:

- useradd creates a new user, passwd sets the password.
- and sudo su changes th users

---

2. Navigate to the home directory

The terminal shows the home directory path.

Command Executed:

cd ~

pwd

Command Output:

Evidence (Screenshot):

Key Learning:

cd ~ takes you directly to the home directory.

---

3. Create a new directory

Command Executed:

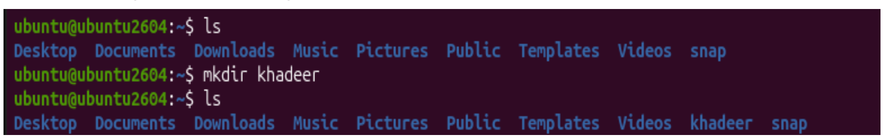
mkdir khadeer

ls

Command Output:

A new directory named khadeer is created.

Evidence (Screenshot):



Key Learning:

- mkdir is used to create directories.
- ls is used to list the directorys

---

4. List the contents of a directory

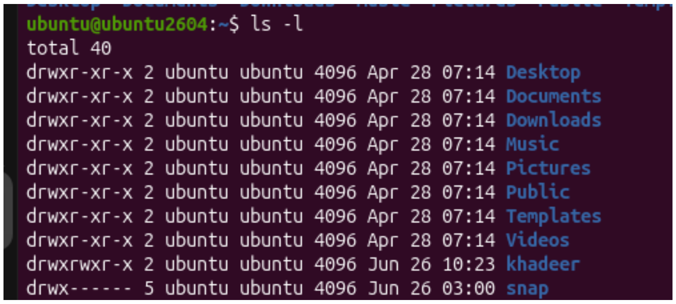
Command Executed:

ls -l

Command Output:

Files and folders are displayed in long format.

Evidence (Screenshot):



Key Learning:

ls -l shows permissions, owner, size, and time.

---

5. Change the current directory

Command Executed:

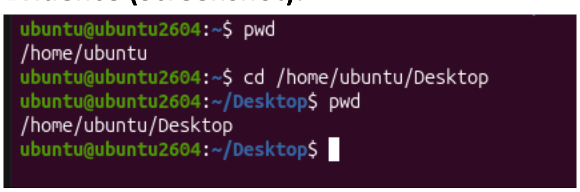
cd /home/unbuntu/Desktop

pwd

Command Output:

Current directory changes to Desktop.

Evidence (Screenshot):



Key Learning:

cd changes the working directory.

---

6. Create a new empty file

Command Executed:

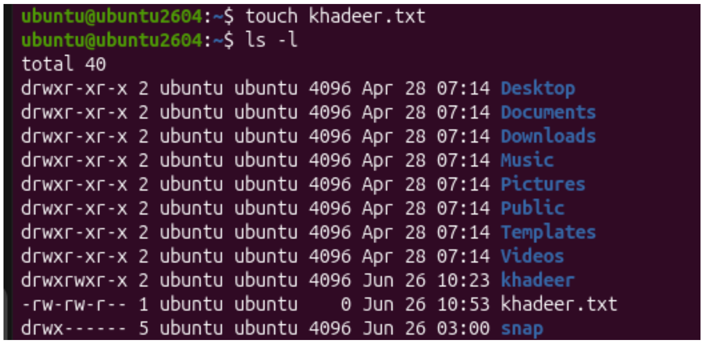
touch khadeer.txt

ls -l

Command Output:

An empty file named khadeer.txt is created.

Evidence (Screenshot):



```
ubuntu@ubuntu2604:~$ touch khadeer.txt
ubuntu@ubuntu2604:~$ ls -l
total 40
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Desktop
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Documents
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Downloads
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Music
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Pictures
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Public
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Templates
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Videos
drwxrwxr-x 2 ubuntu ubuntu 4096 Jun 26 10:23 khadeer
-rw-rw-r-- 1 ubuntu ubuntu    0 Jun 26 10:53 khadeer.txt
drwx------ 5 ubuntu ubuntu 4096 Jun 26 03:00 snap
```

Key Learning:

touch creates empty files.

---

7. View the contents of a file

Command Executed:

cat khadeer.txt

ls

Command Output:

Hi this khadeer

Evidence (Screenshot):

```
ubuntu@ubuntu2604:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos  khadeer  khadeer.txt  snap
ubuntu@ubuntu2604:~$ cat khadeer.txt
Hi this khadeer
```

Key Learning:

cat is used to view small text files.

---

8. Copy a file to another location

Command Executed:

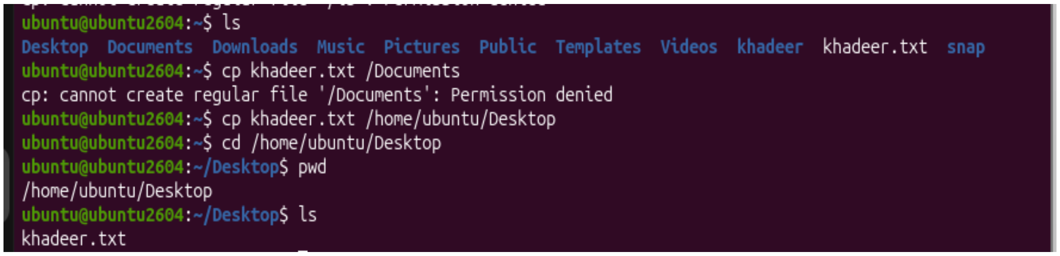
cp khadeer.txt /home/ubuntu/Desktop

ls

Command Output:

The file is copied to the parent directory to Desktop directory.

Evidence (Screenshot):



```
ubuntu@ubuntu2604:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos  khadeer  khadeer.txt  snap
ubuntu@ubuntu2604:~$ cp khadeer.txt /Documents
cp: cannot create regular file '/Documents': Permission denied
ubuntu@ubuntu2604:~$ cp khadeer.txt /home/ubuntu/Desktop
ubuntu@ubuntu2604:~$ cd /home/ubuntu/Desktop
ubuntu@ubuntu2604:~/Desktop$ pwd
/home/ubuntu/Desktop
ubuntu@ubuntu2604:~/Desktop$ ls
khadeer.txt
```

Key Learning:

cp copies files.

---

9. Move a file to another location

Command Executed:

mv file1.txt ../

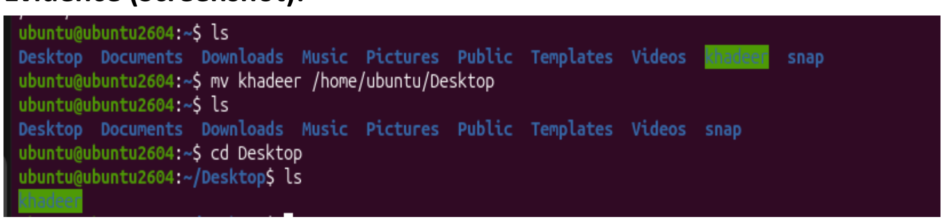
ls

cd Desktop

Command Output:

The file is moved to another location.

Evidence (Screenshot):



```
ubuntu@ubuntu2604:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos  khadeer  snap
ubuntu@ubuntu2604:~$ mv khadeer /home/ubuntu/Desktop
ubuntu@ubuntu2604:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos  snap
ubuntu@ubuntu2604:~$ cd Desktop
ubuntu@ubuntu2604:~/Desktop$ ls
khadeer
```

---

10. Rename a file

Command Executed:

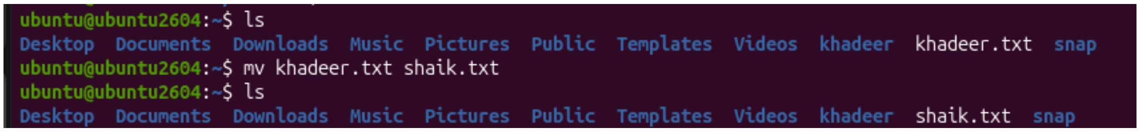
mv khadeer.txt shaik.txt

ls

Command Output:

The file is renamed khadeer.txt to shaik.txt

Evidence (Screenshot):



Key Learning:

- mv is also used to rename files.

---

11. Delete a file

Command Executed:

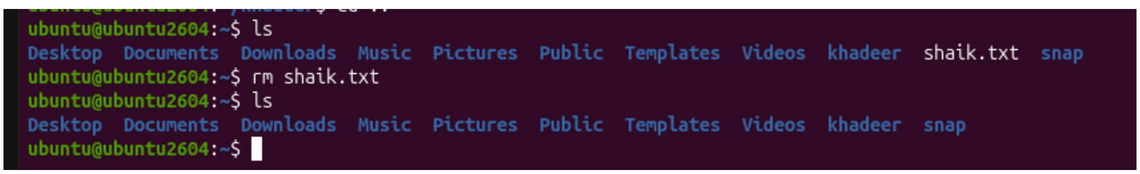
rm khadeer.txt

ls

Command Output:

The file is deleted Sucessfully.

Evidence (Screenshot):



Key Learning:

- rm command used to delete the Files.
- rm deletes files permanently.

---

12. Grant or revoke permissions on a file or directory

Command Executed:

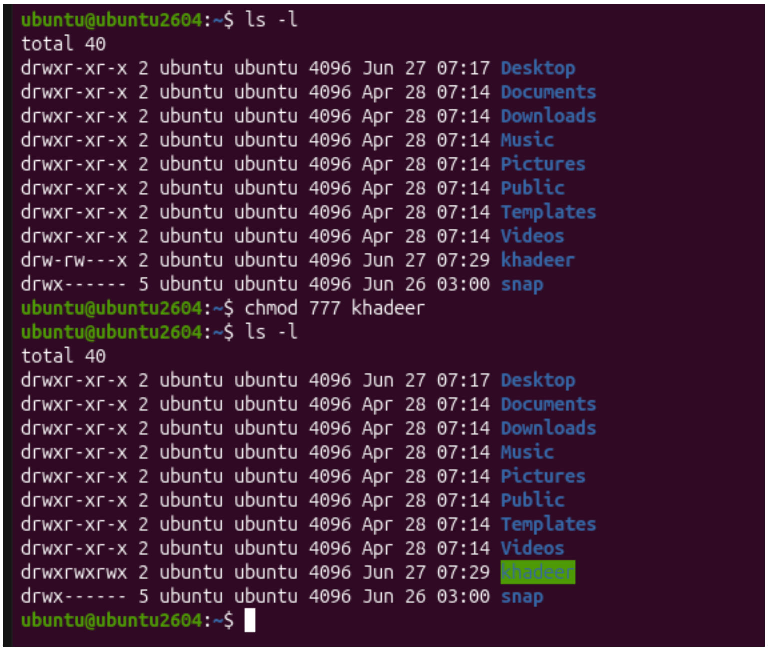
chmod 777 khadeer

ls -l

Command Output:

Permissions are changed.

Evidence (Screenshot):



```
ubuntu@ubuntu2604:~$ ls -l
total 40
drwxr-xr-x 2 ubuntu ubuntu 4096 Jun 27 07:17 Desktop
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Documents
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Downloads
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Music
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Pictures
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Public
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Templates
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Videos
drw-rw---x 2 ubuntu ubuntu 4096 Jun 27 07:29 khadeer
drwx------ 5 ubuntu ubuntu 4096 Jun 26 03:00 snap
ubuntu@ubuntu2604:~$ chmod 777 khadeer
ubuntu@ubuntu2604:~$ ls -l
total 40
drwxr-xr-x 2 ubuntu ubuntu 4096 Jun 27 07:17 Desktop
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Documents
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Downloads
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Music
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Pictures
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Public
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Templates
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Videos
drwxrwxrwx 2 ubuntu ubuntu 4096 Jun 27 07:29 khadeer
drwx------ 5 ubuntu ubuntu 4096 Jun 26 03:00 snap
ubuntu@ubuntu2604:~$
```

Key Learning:

- chmod changes read, write, and execute permissions.
- chmod mays useful some case of permission Changes.

---

13. View the current date and time

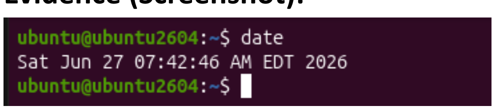
Command Executed:

date

Command Output:

Current date and time are displayed.

Evidence (Screenshot):



```
ubuntu@ubuntu2604:~$ date
Sat Jun 27 07:42:46 AM EDT 2026
ubuntu@ubuntu2604:~$
```

Key Learning:

- date shows system date and time.

---

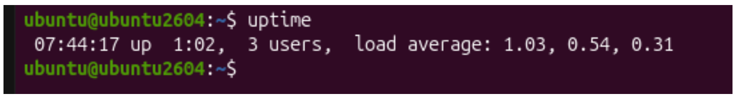
14. Check the system uptime

Command Executed:

uptime

Command Output:
System uptime and load average are shown.

Evidence (Screenshot):



Key Learning:

- uptime shows how long the system has been running.

---

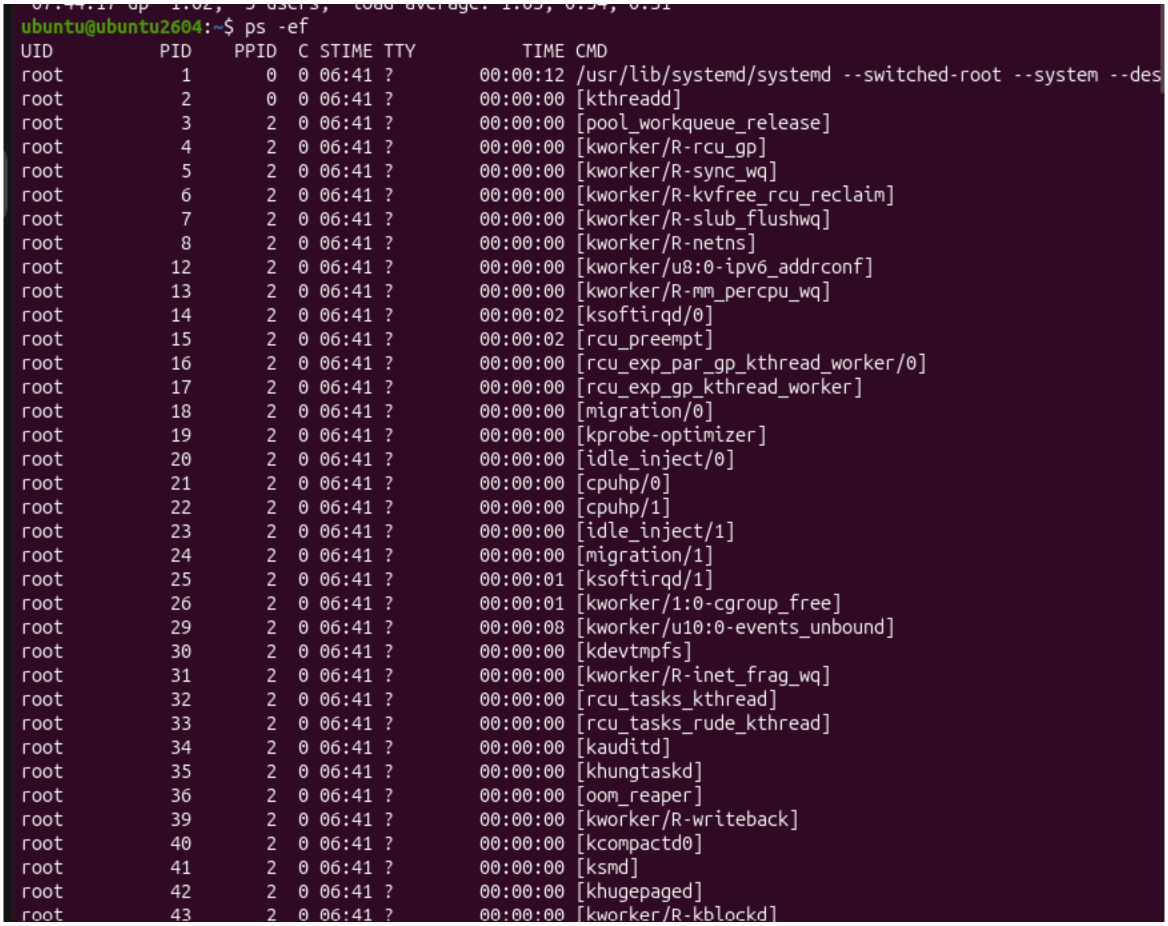
15. View the running processes

Command Executed:

ps -ef

Command Output:
Running processes are listed.

Evidence (Screenshot):



```
ubuntu@ubuntu2604:~$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1       0  0 06:41 ?        00:00:12 /usr/lib/systemd/systemd --switched-root --system --des
root           2       0  0 06:41 ?        00:00:00 [kthreadd]
root           3       2  0 06:41 ?        00:00:00 [pool_workqueue_release]
root           4       2  0 06:41 ?        00:00:00 [kworker/R-rcu_gp]
root           5       2  0 06:41 ?        00:00:00 [kworker/R-sync_wq]
root           6       2  0 06:41 ?        00:00:00 [kworker/R-kvfree_rcu_reclaim]
root           7       2  0 06:41 ?        00:00:00 [kworker/R-slub_flushwq]
root           8       2  0 06:41 ?        00:00:00 [kworker/R-netns]
root          12       2  0 06:41 ?        00:00:00 [kworker/u8:0-ipv6_addrconf]
root          13       2  0 06:41 ?        00:00:00 [kworker/R-mm_percpu_wq]
root          14       2  0 06:41 ?        00:00:02 [ksoftirqd/0]
root          15       2  0 06:41 ?        00:00:02 [rcu_preempt]
root          16       2  0 06:41 ?        00:00:00 [rcu_exp_par_gp_kthread_worker/0]
root          17       2  0 06:41 ?        00:00:00 [rcu_exp_gp_kthread_worker]
root          18       2  0 06:41 ?        00:00:00 [migration/0]
root          19       2  0 06:41 ?        00:00:00 [kprobe-optimizer]
root          20       2  0 06:41 ?        00:00:00 [idle_inject/0]
root          21       2  0 06:41 ?        00:00:00 [cpuhp/0]
root          22       2  0 06:41 ?        00:00:00 [cpuhp/1]
root          23       2  0 06:41 ?        00:00:00 [idle_inject/1]
root          24       2  0 06:41 ?        00:00:00 [migration/1]
root          25       2  0 06:41 ?        00:00:01 [ksoftirqd/1]
root          26       2  0 06:41 ?        00:00:01 [kworker/1:0-cgroup_free]
root          29       2  0 06:41 ?        00:00:08 [kworker/u10:0-events_unbound]
root          30       2  0 06:41 ?        00:00:00 [kdevtmpfs]
root          31       2  0 06:41 ?        00:00:00 [kworker/R-inet_frag_wq]
root          32       2  0 06:41 ?        00:00:00 [rcu_tasks_kthread]
root          33       2  0 06:41 ?        00:00:00 [rcu_tasks_rude_kthread]
root          34       2  0 06:41 ?        00:00:00 [kauditd]
root          35       2  0 06:41 ?        00:00:00 [khungtaskd]
root          36       2  0 06:41 ?        00:00:00 [oom_reaper]
root          39       2  0 06:41 ?        00:00:00 [kworker/R-writeback]
root          40       2  0 06:41 ?        00:00:00 [kcompactd0]
root          41       2  0 06:41 ?        00:00:00 [ksmd]
root          42       2  0 06:41 ?        00:00:00 [khugepaged]
root          43       2  0 06:41 ?        00:00:00 [kworker/R-kblockd]
```

Key Learning:

- ps -ef displays active processes**.**
- ps -ef also displays UID,PID,PPID.

---

16. Kill a running process

Command Executed:

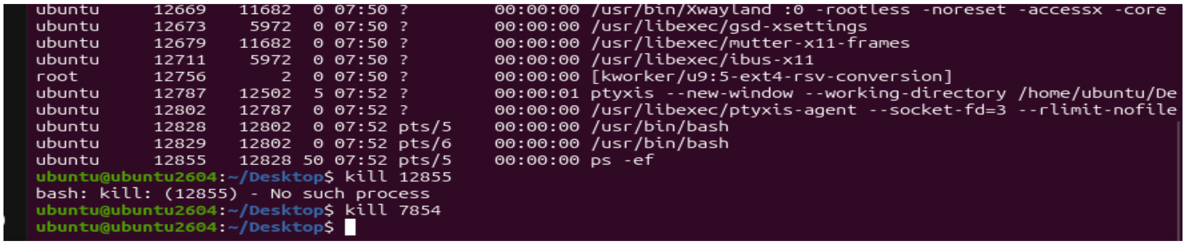
ps -ef

kill 7854

Command Output:

The selected process stops.

Evidence (Screenshot):



Key Learning:

- kill stops a process using its PID.

---

17. Install a package using the package manager

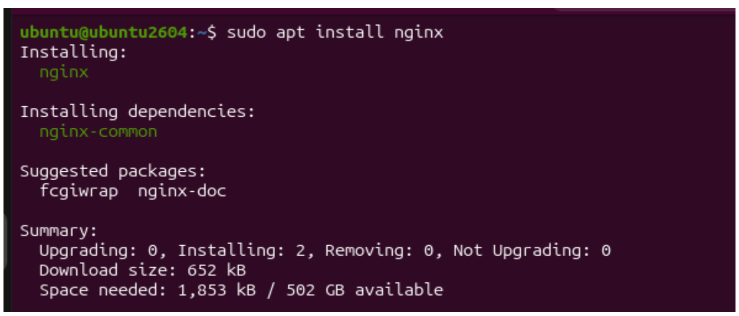
Command Executed:

sudo apt install nginx

Command Output:

Package gets installed.

Evidence (Screenshot):



Key Learning:

apt installs packages on Debian-based systems.

---

18. Update the system packages

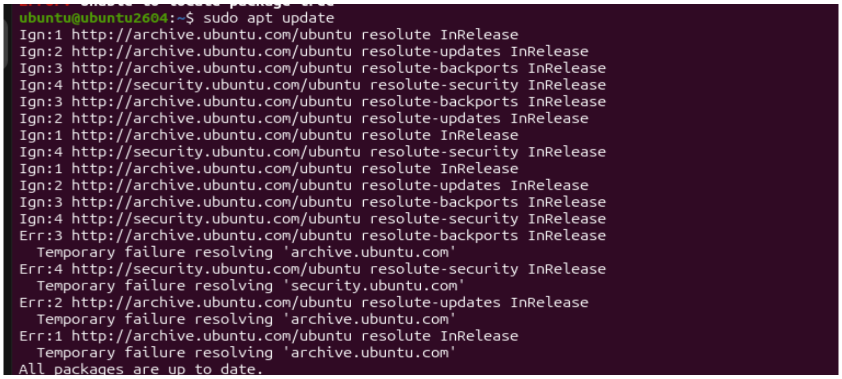
Command Executed:

sudo apt update

Command Output:

System package lists are updated.

Evidence (Screenshot):



```
ubuntu@ubuntu2604:~$ sudo apt update
Ign:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Ign:2 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Ign:3 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Ign:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
Ign:3 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Ign:2 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Ign:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Ign:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
Ign:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Ign:2 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Ign:3 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Ign:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
Err:3 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
  Temporary failure resolving 'archive.ubuntu.com'
Err:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
  Temporary failure resolving 'security.ubuntu.com'
Err:2 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
  Temporary failure resolving 'archive.ubuntu.com'
Err:1 http://archive.ubuntu.com/ubuntu resolute InRelease
  Temporary failure resolving 'archive.ubuntu.com'
All packages are up to date.
```

Key Learning:

Updating keeps the system secure and current**.**

---

19. Create a symbolic link

A symbolic link, or symlink, is like a shortcut that points to another file or folder instead of containing the actual data itself.

Command Executed:

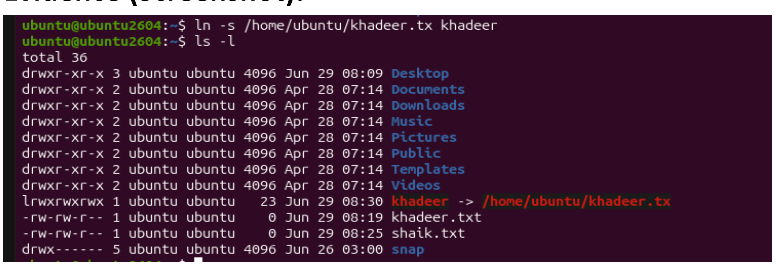
ln -s /home/ubuntu/khadeer.txt khadeer

ls -l

Command Output:

A symbolic link is created.

Evidence (Screenshot):



```
ubuntu@ubuntu2604:~$ ln -s /home/ubuntu/khadeer.tx khadeer
ubuntu@ubuntu2604:~$ ls -l
total 36
drwxr-xr-x 3 ubuntu ubuntu 4096 Jun 29 08:09 Desktop
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Documents
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Downloads
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Music
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Pictures
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Public
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Templates
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Videos
lrwxrwxrwx 1 ubuntu ubuntu   23 Jun 29 08:30 khadeer -> /home/ubuntu/khadeer.tx
-rw-rw-r-- 1 ubuntu ubuntu    0 Jun 29 08:19 khadeer.txt
-rw-rw-r-- 1 ubuntu ubuntu    0 Jun 29 08:25 shaik.txt
drwx------ 5 ubuntu ubuntu 4096 Jun 26 03:00 snap
```

Key Learning:

ln -s creates a soft link.

---

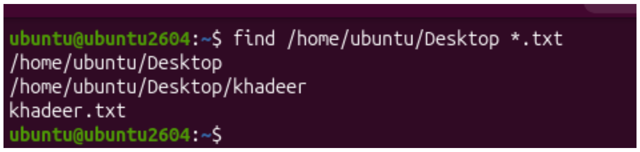
20. Search for files using the find command

Command Executed:

find /home/ubuntu/Desktop *.txt

Command Output:
Matching files are displayed.

Evidence (Screenshot):



```
ubuntu@ubuntu2604:~$ find /home/ubuntu/Desktop *.txt
/home/ubuntu/Desktop
/home/ubuntu/Desktop/khadeer
khadeer.txt
ubuntu@ubuntu2604:~$
```

Key Learning:
find searches files and folders.

21- Compress and decompress files using tar.

tar is used to bundle files and folders into one archive, and it can also compress or extract that archive in one step.

Command Executed:

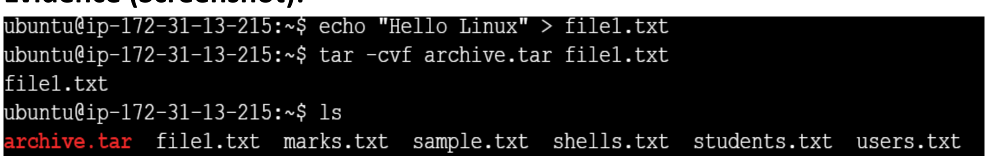
echo “Hello Linux” > file1.txt

-tar -cvf archive.tar file1.txt

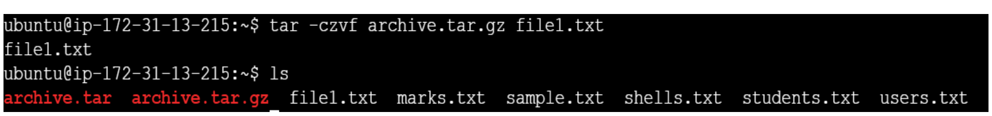
tar -czvf archive.tar.gz file.txt

Command Output:
Archive is created and extracted.

Evidence (Screenshot):



```
ubuntu@ip-172-31-13-215:~$ echo "Hello Linux" > file1.txt
ubuntu@ip-172-31-13-215:~$ tar -cvf archive.tar file1.txt
file1.txt
ubuntu@ip-172-31-13-215:~$ ls
archive.tar  file1.txt  marks.txt  sample.txt  shells.txt  students.txt  users.txt
```



```
ubuntu@ip-172-31-13-215:~$ tar -czvf archive.tar.gz file1.txt
file1.txt
ubuntu@ip-172-31-13-215:~$ ls
archive.tar  archive.tar.gz  file1.txt  marks.txt  sample.txt  shells.txt  students.txt  users.txt
```

Key Learning:
tar is used for archiving and extraction.

---

22. Monitor system resources with top or htop

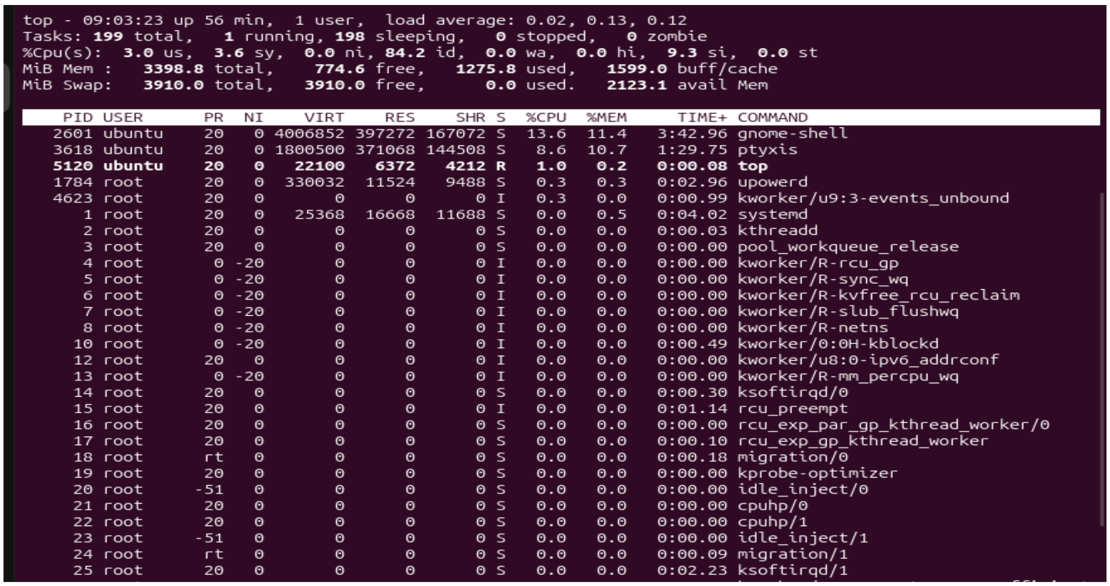
Command Executed:

top

Command Output:

Live CPU, memory, and process usage shown.

Evidence (Screenshot):



```
top - 09:03:23 up 56 min,  1 user,  load average: 0.02, 0.13, 0.12
Tasks: 199 total,   1 running, 198 sleeping,   0 stopped,   0 zombie
%Cpu(s):  3.0 us,  3.6 sy,  0.0 ni, 84.2 id,  0.0 wa,  0.0 hi,  9.3 si,  0.0 st
MiB Mem :   3398.8 total,    774.6 free,   1275.8 used,   1599.0 buff/cache
MiB Swap:   3910.0 total,   3910.0 free,      0.0 used.   2123.1 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   2601 ubuntu    20   0 4006852 397272 167072 S  13.6  11.4   3:42.96 gnome-shell
   3618 ubuntu    20   0 1800500 371068 144508 S   8.6  10.7   1:29.75 ptyxis
   5120 ubuntu    20   0   22100   6372   4212 R   1.0   0.2   0:00.08 top
   1784 root      20   0  330032  11524   9488 S   0.3   0.3   0:02.96 upowerd
   4623 root      20   0       0      0      0 I   0.3   0.0   0:00.99 kworker/u9:3-events_unbound
      1 root      20   0   25368  16668  11688 S   0.0   0.5   0:04.02 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.03 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_release
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
      7 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-slub_flushwq
      8 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-netns
     10 root       0 -20       0      0      0 I   0.0   0.0   0:00.49 kworker/0:0H-kblockd
     12 root      20   0       0      0      0 I   0.0   0.0   0:00.00 kworker/u8:0-ipv6_addrconf
     13 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-mm_percpu_wq
     14 root      20   0       0      0      0 S   0.0   0.0   0:00.30 ksoftirqd/0
     15 root      20   0       0      0      0 I   0.0   0.0   0:01.14 rcu_preempt
     16 root      20   0       0      0      0 S   0.0   0.0   0:00.00 rcu_exp_par_gp_kthread_worker/0
     17 root      20   0       0      0      0 S   0.0   0.0   0:00.10 rcu_exp_gp_kthread_worker
     18 root      rt   0       0      0      0 S   0.0   0.0   0:00.18 migration/0
     19 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kprobe-optimizer
     20 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 idle_inject/0
     21 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/0
     22 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/1
     23 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 idle_inject/1
     24 root      rt   0       0      0      0 S   0.0   0.0   0:00.09 migration/1
     25 root      20   0       0      0      0 S   0.0   0.0   0:02.23 ksoftirqd/1
```

Key Learning:

top helps monitor performance.

---

23. Create and manage user groups

Command Executed:

sudo groupadd developers

getent group developers

sudo usermod -aG developers ubuntu

cat /etc/group

Command Output:

Group created and user added.

Evidence (Screenshot):

```
ubuntu@ip-172-31-13-215:~$ sudo groupadd developers
ubuntu@ip-172-31-13-215:~$ getent group developers
developers:x:1002:
ubuntu@ip-172-31-13-215:~$ sudo usermod -aG developers ubuntu
```

```
ssl-cert:x:109:
mysql:x:110:
tomcat:x:1001:
developers:x:1002:ubuntu
```

Key Learning:
Groups help manage permissions.

---

24. Set up SSH passwordless authentication

Command Executed:

ssh-keygen -t ed25519

ssh-copy-id user@server

Command Output:
SSH key pair created and copied to remote server.

Evidence (Screenshot):
Screenshot of key generation and test login.

Key Learning:
SSH keys allow secure passwordless login.

---

25. Monitor log files using tail or grep

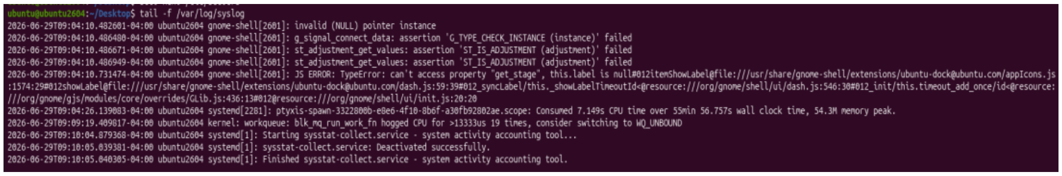
Command Executed:

tail -f /var/log/syslog

grep "error" /var/log/syslog

Command Output:
Logs are shown live or filtered.

Evidence (Screenshot):



Key Learning:

tail and grep are useful for troubleshooting.

---

26. Set up a web server

Command Executed:

sudo apt install apache2

sudo systemctl enable apache2

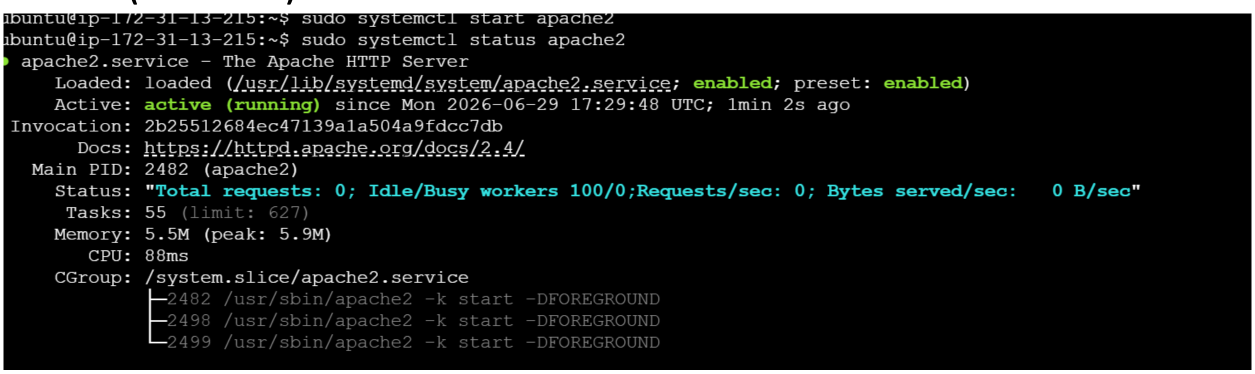
sudo systemctl start apache2

sudo systemctl status apache2

Command Output:

Apache is installed and running.

Evidence (Screenshot):



---

27. Configure and secure MySQL database

Command Executed:

sudo apt install mysql-server

sudo mysql_secure_installation

mysql -u root -u

Command Output:
MySQL installed and secured.

Evidence (Screenshot):

```
mysql> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
4 rows in set (0.04 sec)
```

Key Learning:
mysql_secure_installation improves database security.

---

28. Set up Apache Tomcat application server

Command Executed:

sudo apt update

sudo apt install openjdk-21-jre-headless

java -version

Evidence (Screenshot):

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
openjdk version "21.0.11" 2026-04-21
OpenJDK Runtime Environment (build 21.0.11+10-1-26.04.2-Ubuntu)
OpenJDK 64-Bit Server VM (build 21.0.11+10-1-26.04.2-Ubuntu, mixed mode, sharing)
ubuntu@ip-172-31-13-215:~$ java -version
openjdk version "21.0.11" 2026-04-21
OpenJDK Runtime Environment (build 21.0.11+10-1-26.04.2-Ubuntu)
OpenJDK 64-Bit Server VM (build 21.0.11+10-1-26.04.2-Ubuntu, mixed mode, sharing)
ubuntu@ip-172-31-13-215:~$
```

---

29. Create a service file for Apache Tomcat

Command Executed:

sudo nano /etc/systemd/system/tomcat.service

Service File Content:

text

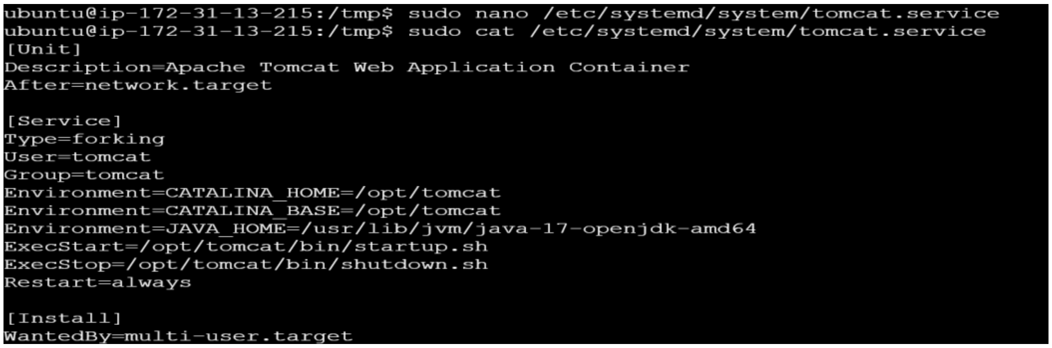
```
[Unit]
Description=Apache Tomcat Web Application Container
After=network.target

[Service]
Type=forking
User=tomcat
Group=tomcat
Environment=CATALINA_HOME=/opt/tomcat
Environment=CATALINA_BASE=/opt/tomcat
Environment=JAVA_HOME=/usr/lib/jvm/ openjdk-17-jre-headless
ExecStart=/opt/tomcat/bin/startup.sh
ExecStop=/opt/tomcat/bin/shutdown.sh
Restart=on-failure

[Install]
WantedBy=multi-user.target
```

Command Output:
Service file saved successfully.

Evidence (Screenshot):



```
ubuntu@ip-172-31-13-215:/tmp$ sudo nano /etc/systemd/system/tomcat.service
ubuntu@ip-172-31-13-215:/tmp$ sudo cat /etc/systemd/system/tomcat.service
[Unit]
Description=Apache Tomcat Web Application Container
After=network.target

[Service]
Type=forking
User=tomcat
Group=tomcat
Environment=CATALINA_HOME=/opt/tomcat
Environment=CATALINA_BASE=/opt/tomcat
Environment=JAVA_HOME=/usr/lib/jvm/java-17-openjdk-amd64
ExecStart=/opt/tomcat/bin/startup.sh
ExecStop=/opt/tomcat/bin/shutdown.sh
Restart=always

[Install]
WantedBy=multi-user.target
```

Key Learning:

systemd service files allow Tomcat to be managed with systemctl.

---

30. Start Tomcat using systemctl

Command Executed:

sudo systemctl daemon-reload

sudo systemctl enable tomcat

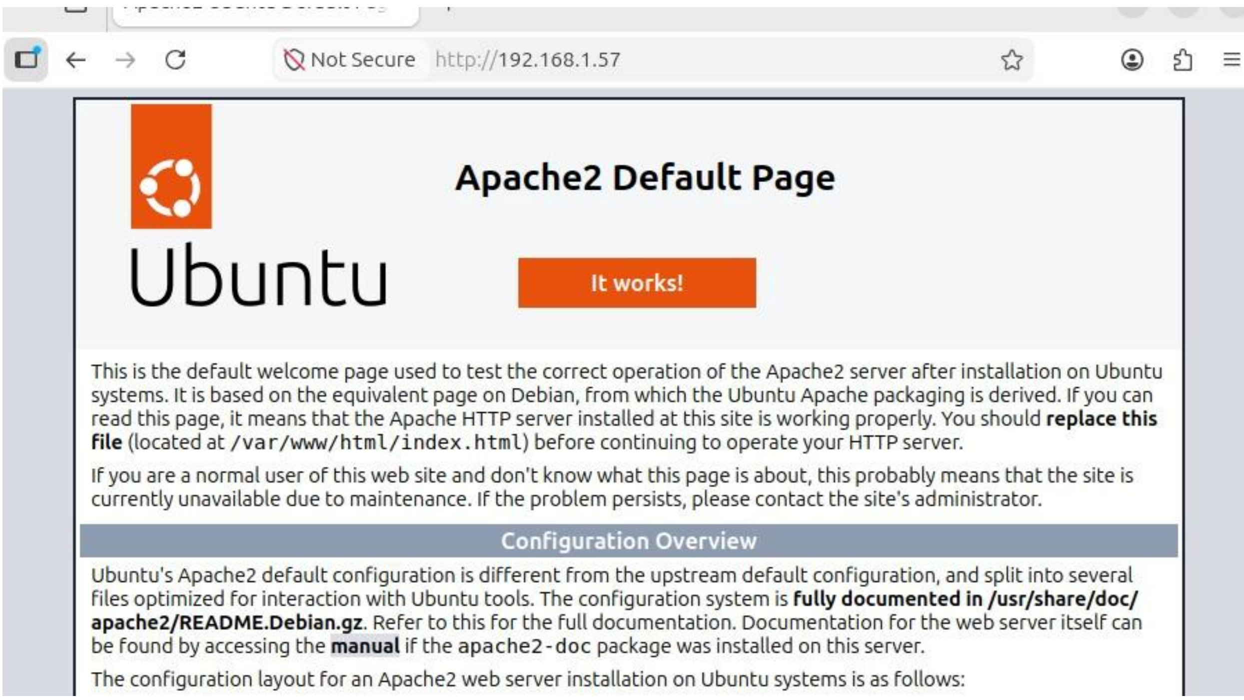
sudo systemctl start tomcat

sudo systemctl status tomcat

Command Output:

Tomcat service starts successfully.

Evidence (Screenshot):



---

31. Print specific columns from a delimited file

Command Executed:

awk -F, '{print $1, $7}' /etc/passwd

Command Output:

Selected columns are printed.

Evidence (Screenshot):

```
ubuntu@ip-172-31-13-215:/tmp$ awk -F':' '{print $1, $7}' /etc/passwd
root /bin/bash
daemon /usr/sbin/nologin
bin /usr/sbin/nologin
sys /usr/sbin/nologin
sync /bin/sync
games /usr/sbin/nologin
man /usr/sbin/nologin
lp /usr/sbin/nologin
mail /usr/sbin/nologin
news /usr/sbin/nologin
uucp /usr/sbin/nologin
proxy /usr/sbin/nologin
www-data /usr/sbin/nologin
backup /usr/sbin/nologin
```

Key Learning:

AWK is useful for extracting columns.

---

32. Filter and print lines based on a pattern

Command Executed:

grep "root" /etc/passwd

Command Output:
Only matching lines are printed.

Evidence (Screenshot):

```
ubuntu@ip-172-31-13-215:/etc$ grep "root" /etc/passwd
root:x:0:0:root:/root:/bin/bash
```

Key Learning:
grep searches for patterns.

---

33. Calculate and print average, sum, or statistics

Command Executed:

awk '{sum += $1} END {print "Sum =", sum}' numbers.txt

ls

Command Output:
Sum and average are displayed.

Evidence (Screenshot):

```
ubuntu@ip-172-31-13-215:/tmp$ awk '{sum += $1} END {print "Sum =", sum}' numbers.txt
Sum = 150
```

Key Learning:
AWK can perform arithmetic on columns.

---

34. Perform string manipulation

Command Executed:

echo "hello world" | tr '[:lower:]' '[:upper:]'

Command Output:
Text is printed in uppercase.

Evidence (Screenshot):

```
ubuntu@ip-172-31-13-215:~$ echo "hello world" | tr '[:lower:]' '[:upper:]'
HELLO WORLD
```

---

35. Count the occurrences of a specific pattern

Command Executed:

grep -c "root" /etc/passwd

Command Output:
Number of matching lines is shown.

Evidence (Screenshot):

```
ubuntu@ip-172-31-13-215:~$ grep -c "root" /etc/passwd
1
```

Key Learning:
grep -c counts matching lines.

---

36. Sort lines based on a specific field

Command Executed:

sort -t: -k3,3n /etc/passwd

Command Output:
Rows are sorted by the second field.

Evidence (Screenshot):

```
ubuntu@ip-172-31-13-215:~$ sort -t: -k3,3n /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
```

Key Learning:
sort helps arrange data by columns.

---

37. Merge multiple files based on a common field

Command Executed:

cat > students.txt << EOF

101 Alice

102 Bob

103 Charlie

EOF

cat > marks.txt << EOF

101 85

102 90

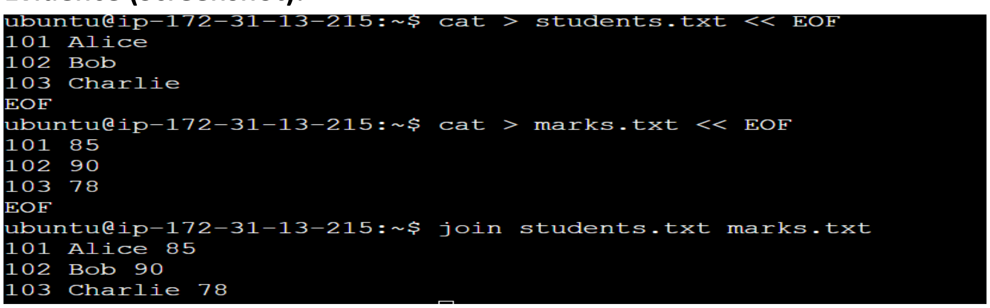
103 78

EOF

join students.txt marks.txt

Command Output:
Combined data is displayed.

Evidence (Screenshot):



Key Learning:
join combines files with common fields.

---

38. Substitute text in a file

Command Executed:

echo "Linux is good. Linux is fast." > sample.txt

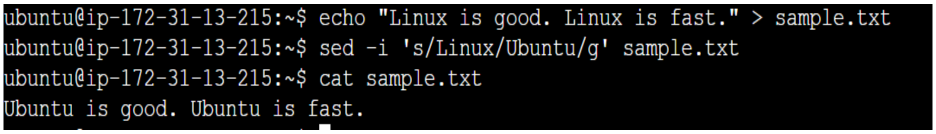
sed -i 's/Linux/Ubuntu/g' sample.txt

cat sample.txt

Command Output:

Text is replaced in output.

Evidence (Screenshot):



Key Learning:

sed is used for search and replace.

---

39. Delete specific lines based on a pattern or line number

Command Executed:

grep -v "root" /etc/passwd

sed '2,5d' /etc/passwd

Command Output:

Delete lines 2 through 5

Evidence (Screenshot):

Key Learning:

sed can delete lines by number.

---

40. Append or insert text before or after a pattern

Command Executed:

cat > sample.txt << EOF

apple

banana

orange

EOF

sed '/banana/a\grapes' sample.txt

Command Output:
A line is inserted before the matched pattern.

Evidence (Screenshot):

```
ubuntu@ip-172-31-13-215:~$ cat > sample.txt << EOF
apple
banana
orange
EOF
ubuntu@ip-172-31-13-215:~$ sed '/banana/a\grapes' sample.txt
apple
banana
grapes
orange
```

Key Learning:
sed can add text around a pattern.

---

41. Print only specific lines

Command Executed:

sed -n '2,5p' /etc/passwd

Command Output:
Only selected lines are printed.

Evidence (Screenshot):

```
ubuntu@ip-172-31-13-215:~$ sed -n '1,5p' /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
```

Key Learning:
sed -n prints only requested lines.

---

42. Copy file from Linux to Windows machine

Command Executed:

bash

scp file.txt windowsuser@windows-ip:/path/

Command Output:
File transferred successfully.

Evidence (Screenshot):
Screenshot of transfer command.

Key Learning:
scp or pscp is used to transfer files between Linux and Windows.stackoverflow+1
